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# =====
# STAGE 1 - CONTAINER 1
# DATASET OVERVIEW
# =====

print("DATASET OVERVIEW")
print("=" * 50)
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv("/content/Dataset(1).csv")

print("Dataset loaded successfully!")
print("Shape:", df.shape)
rows, columns = df.shape

print(f"Number of records : {rows:,}")
print(f"Number of columns : {columns}")

print("\nDataset memory usage:")
memory_mb = df.memory_usage(deep=True).sum() / (1024 ** 2)
print(f"{memory_mb:.2f} MB")

print("\nIndex information:")
print(f"Index type      : {type(df.index).__name__}")
print(f"Index start     : {df.index.min()}")
print(f"Index end       : {df.index.max()}")
```

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DATASET OVERVIEW
=====
Dataset loaded successfully!
Shape: (250000, 27)
Number of records : 250,000
Number of columns : 27

Dataset memory usage:
344.52 MB

Index information:
Index type      : RangeIndex
Index start     : 0
Index end       : 249999
```

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# =====
# STAGE 1 - CONTAINER 2
# COLUMN AND DATA-TYPE PROFILE
# =====

print("COLUMN AND DATA-TYPE PROFILE")
print("=" * 70)

profile = pd.DataFrame({
    "Column": df.columns,
    "Data Type": df.dtypes.astype(str).values,
    "Non-Null Count": df.notna().sum().values,
    "Unique Values": df.nunique(dropna=True).values
})

display(profile)
```

COLUMN AND DATA-TYPE PROFILE

	Column	Data Type	Non-Null Count	Unique Values
0	Transaction ID	object	250000	250000
1	time_stamp	object	245357	189975
2	Transaction_Type	object	244939	55
3	merchant category	object	244936	91
4	amount_INR	object	245364	19230
5	Transaction Status	object	244959	35
6	sender_agegroup	object	245437	58
7	receiver_age_group	object	245428	57
8	Sender State	object	245428	91
9	sender bank	object	245434	81
10	Receiver_Bank	object	245443	83
11	device type	object	245415	36
12	Network Type	object	245422	46
13	Fraud Flag	object	245856	6
14	hour of day	object	245955	28
15	Day Of Week	object	245921	11
16	is weekend	object	245923	6
17	battery level %	object	243699	99
18	App Version	object	244966	81
19	device storage free gb	object	244919	2544
20	screen brightness %	object	244902	94
21	Device Language	object	244964	84
22	GPS accuracy m	object	244926	1113
23	nearby wifi count	object	244913	19
24	session duration sec	object	244917	457
25	notifications last hour	object	244912	22
26	merchant distance km	object	244918	1929

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# =====
# STAGE 1 - CONTAINER 3
# RECORD INSPECTION
# =====

print("FIRST 5 RECORDS")
print("=" * 50)
display(df.head())

print("\nLAST 5 RECORDS")
print("=" * 50)
display(df.tail())

print("\n5 RANDOM RECORDS")
print("=" * 50)
display(df.sample(5, random_state=42))
```

FIRST 5 RECORDS

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	Transaction ID	time_stamp	Transaction_Type	merchant category	amount_INR	Transaction Status	sender_agegroup	receiver_age_group	Se S
0	TXN0000000001	10/8/2024 15:17	P2P	Entertainment	868	SUCCESS	26-35	18-25	I
1	TXN0000000002	2024/04/11 06:56:00	P2M	Grocery	1011	SUCCESS	26-35	26-35	Pra
2	TXN0000000003	4/2/2024 13:27	P2P	Grocery	477	SUCCESS	26-35	36-45	Karne
3	TXN0000000004	1/7/2024 10:09	P2P	Fuel	2784	SUCCESS	26-35	26-35	I
4	TXN0000000005	1/23/2024 19:04	P2P	Shopping	990	SUCCESS	26-35	18-25	I

LAST 5 RECORDS

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	Transaction ID	time_stamp	Transaction_Type	merchant category	amount_INR	Transaction Status	sender_agegroup	receiver_age_group	
249995	TXN0000249996	11/8/2024 22:41	Recharge	Food	373	SUCCESS	36-45	26-35	
249996	TXN0000249997	12/15/2024 2:58	P2P	Utilities	2025	SUCCESS	36-45	26-35	
249997	TXN0000249998	11/27/2024 16:33	P2P	Food	468	SUCCESS	26-35	18-25	
249998	TXN0000249999	1/5/2024 13:31	Recharge	Healthcare	284	SUCCESS	18-25	36-45	
249999	TXN0000250000	1/17/2024 15:23	P2P	Entertainment	531	SUCCESS	18-25	26-35	

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# =====
# STAGE 1 - CONTAINER 4
# DATA QUALITY INSPECTION
# =====
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```
print("DATA QUALITY INSPECTION")
print("=" * 60)

# Missing values
missing = df.isnull().sum()

# Duplicate rows
duplicates = df.duplicated().sum()

# Total missing cells
total_missing = missing.sum()

print(f"Total missing cells : {total_missing:,}")
print(f"Duplicate rows      : {duplicates:,}")

print("\nMissing values by column:")
missing_table = pd.DataFrame({
    "Column": missing.index,
    "Missing Values": missing.values
})

display(missing_table)

# Explicit conclusion
print("\nDATA QUALITY RESULT")
print("=" * 60)

if total_missing == 0:
    print("✓ No missing values were found.")

if duplicates == 0:
    print("✓ No duplicate rows were found.")
```

DATA QUALITY INSPECTION

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Total missing cells : 124,747

Duplicate rows : 0

Missing values by column:

	Column	Missing Values
0	Transaction ID	0
1	time_stamp	4643
2	Transaction_Type	5061
3	merchant category	5064
4	amount_INR	4636
5	Transaction Status	5041
6	sender_agegroup	4563
7	receiver_age_group	4572
8	Sender State	4572
9	sender bank	4566
10	Receiver_Bank	4557
11	device type	4585
12	Network Type	4578
13	Fraud Flag	4144
14	hour of day	4045
15	Day Of Week	4079
16	is weekend	4077
17	battery level %	6301
18	App Version	5034
19	device storage free gb	5081
20	screen brightness %	5098
21	Device Language	5036
22	GPS accuracy m	5074
23	nearby wifi count	5087
24	session duration sec	5083
25	notifications last hour	5088
26	merchant distance km	5082

DATA QUALITY RESULT

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✓ No duplicate rows were found.

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# =====
# STAGE 1 - CONTAINER 5
# INITIAL NUMERICAL PROFILE
# =====

print("INITIAL NUMERICAL PROFILE")
print("=" * 70)

# The current dataset contains numerical fields stored as text/object
# values because several fields contain blank entries. For Stage 1,
# we only profile these fields; the original dataframe is not modified.

numeric_data = pd.DataFrame(index=df.index)

for column in df.columns:
    converted = pd.to_numeric(df[column], errors="coerce")

    if converted.notna().any():
        numeric_data[column] = converted
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numeric_columns = numeric_data.columns.tolist()

print(f"Number of numerical columns identified: {len(numeric_columns)}")

print("\nNumerical columns:")
for i, column in enumerate(numeric_columns, start=1):
    print(f"{i}. {column}")

print("\nBasic numerical summary:")

if len(numeric_columns) > 0:
    display(numeric_data[numeric_columns].describe().T)
else:
    print("No numerical columns were identified for profiling.")

print("\nNumerical profiling note:")
print("Numeric-looking fields were converted only in a temporary profiling table.")
print("The original dataframe remains unchanged for the next stage.")

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INITIAL NUMERICAL PROFILE

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Number of numerical columns identified: 12

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Numerical columns:

1. amount_INR
2. Fraud Flag
3. hour of day
4. is weekend
5. battery level %
6. device storage free gb
7. screen brightness %
8. GPS accuracy m
9. nearby wifi count
10. session duration sec
11. notifications last hour
12. merchant distance km

Basic numerical summary:

	count	mean	std	min	25%	50%	75%	max
amount_INR	230681.0	1288.805363	1829.882227	10.0	284.0	615.00	1554.00	42099.0
Fraud Flag	236088.0	0.001902	0.043569	0.0	0.0	0.00	0.00	1.0
hour of day	242976.0	15.679664	10.931996	-1.0	11.0	16.00	19.00	99.0
is weekend	239795.0	0.290411	0.506763	-1.0	0.0	0.00	1.00	2.0
battery level %	240510.0	52.986836	30.496110	-10.0	28.0	52.00	77.00	150.0
device storage free gb	241200.0	1642.065501	12207.473086	-5.0	63.7	129.10	194.50	99999.0
screen brightness %	241175.0	55.408322	29.430525	-10.0	32.0	55.00	78.00	150.0
GPS accuracy m	241272.0	1543.373568	12261.654160	-5.0	8.6	13.50	21.20	99999.0
nearby wifi count	241358.0	19.322591	122.841771	-1.0	3.0	4.00	5.00	999.0
session duration sec	241242.0	14812.520498	120468.769729	-20.0	41.0	71.00	113.00	999999.0
notifications last hour	241202.0	19.943106	121.314758	-1.0	3.0	5.00	6.00	999.0
merchant distance km	241226.0	1527.829169	12252.936616	-5.0	1.3	2.61	4.61	99999.0

Numerical profiling note:

Numeric-looking fields were converted only in a temporary profiling table.
The original dataframe remains unchanged for the next stage.

Start coding or [generate](#) with AI.

Stage 1 Conclusion

The UPI transaction dataset was successfully loaded and inspected using the same Stage 1 workflow used for the project.

The initial inspection established the dataset dimensions, complete column structure, data types, record samples, unique-value information, missing-value status, duplicate-row status and numerical-field profile.

The current dataset contains **250,000 records across 27 columns**. The initial data-quality inspection identified **124,747 missing cells** and **0 duplicate rows**.